Cortical Decoders Trained on Other Individuals Age Slower Than Your Own

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Abstract

With adapting brain–computer interface (BCI) technology, neural decoding algorithms are one of the most essential parts of being able to understand people's brains. Decoders must work for months, and recalibrating them per subject has high subject costs. A decoder fit to a singular animal's cortex drifts, losing accuracy over days, and usually the remedy is recalibrating on fresh data from that same subject. We report a different kind of robustness: a decoder built from one other individual, or pooled from a set of other individuals, ages more slowly than the subject's own. On the open BraiDyn-BC cohort (25 mice, 44 Allen-atlas cortical parcels, a $\sim$15-day operant protocol) we decode four task events and measure, per mouse, the AUC lost over two weeks by its own early-week decoder against one pooled over the other 24 mice and one built from a single other mouse. The personal decoder ages more by $+0.017$ AUC ($p = 5 \times 10^{-5}$, 73 of 100 mouse–event pairs). This also is not a data-volume effect: at a matched training-event count a decoder from a single other mouse still ages less. Hence we conclude this is rather an effect of not overfitting an individual's drift-prone idiosyncrasies. It also survives a 1-D CNN and a GRU. The effect holds past this cohort and recording modality. On an independent two-photon dataset from a different laboratory – Allen Visual Behavior, 23 mice, single-plane VISp populations decoding a stimulus change – the same session-held-out protocol gives $+0.023$ AUC ($p = 0.003$, 74% of mice), and the count-matched control reproduces there as well (self $-$ one other mouse $= +0.011$ AUC). The early-week accuracy cost of decoding across individuals is repaid as late-week stability, which suggests population priors as an alternative to per-subject recalibration. A second offering is the session-held-out protocol under which all of this is measured: because only the within-subject arm can share a recording session with its test data, a trial-level split inflates that arm alone, and holding out whole sessions is what allows this form of robustness to be reported reliably. Code and a streamed-feature pipeline are released.

1 Introduction

Neural decoding is one of the most essential ways to understand neural interfaces and create BCI technology, and a decoder that is deployed is a decoder that has to keep working: over months, on the same individual, without fresh labelled data arriving to prop it up. Representational drift stands directly against that. The reorganization of an animal's neural code over days with fixed behavior is now documented across sensory, motor, and association cortex and in the hippocampus [18, 3, 20]. This has a stark impact on neural decoding: a decoder trained on an animal today degrades on that same animal weeks later. The dominant response is recalibration, periodically refitting the decoder on fresh within-subject data [2, 5], which requires ongoing labelled data from the individual and treats each subject in isolation.

The code's cross-individual structure offers an alternate route to temporal robustness, and the field already relies on it. Cortex-wide behavioral representations are considerably shared among individuals [11, 7, 8]; latent dynamics are preserved across animals well enough that a decoder fit to one transfers to another [17]; and pooled decoders already read out behavior from unseen individuals [21, 12, 13]. Methods such as Meta-AlignNN [10] are built on that premise. While people currently know that conservation across individuals is established, and that pooling is known to work at a fixed time, every one of these results is measured at a fixed time. We offer pooling as a solution to a new problem: the fact that it resists drift over time.

Our core discovery is that a decoder that never saw the subject cannot fit to that subject's drift-prone idiosyncrasies, so it should age more slowly, and the way we measure that difference is as an asymmetry between the two arms. This prediction, to our knowledge, has not been examined.

We test it on BraiDyn-BC [6], a widefield calcium-imaging resource whose $\sim$15-day operant protocol and shared 44-region Allen parcellation make within-subject and cross-subject decoders directly comparable across time. Our contributions:

- **Pooling resists drift.** Across a two-week period the population-pooled decoder loses less accuracy than the individual's own, with a paired asymmetry of $+0.017$ AUC ($p = 5\times10^{-5}$, 73 of 100 pairs; Section 4.1). The size of this effect varies among the four events, but those are chained within a trial and have highly differing counts, so we treat the pooled estimate as the result and leave the per-event ranking uninterpreted.
- **It is not a data-volume artifact.** At a matched training-event count a decoder trained on one other mouse ages less than one trained on the subject's own, by $+0.010$ AUC on average. The gain therefore comes from training on individuals other than the subject, not from the larger training set pooling usually brings, and training on many other mice rather than one lowers aging further on all four events (Section 4.2).
- **It replicates on an independent dataset.** On the Allen Visual Behavior two-photon cohort the same protocol gives $+0.023$ AUC ($p = 0.003$, 74% of 23 mice), with the count-matched ordering also reproduced (Section 4.4).
- **Session-level validation is load-bearing.** As trials within a recording session are dependent, a trial-level split inflates the personal decoder but not the pooled one, one-sidedly inflating the asymmetry. We report that inflation for each event; on lever pull it accounts for the entire apparent effect. For longitudinal decoding comparisons we recommend session-held-out validation as a default (Section 4.3).

2 Related work

Representational drift and its mitigation. Drift is characterized within individual subjects [18, 3, 20, 16], and most approaches to mitigating it also operate within the subject. These include manifold stabilization [2], supervised and self-calibrating recalibration [5, 15], and adversarial or nonlinear alignment to the subject's own reference day [4, 14]. Each method uses later data from the same subject to stabilize the decoder. We instead ask whether data from other subjects can provide that stability.

Cross-subject decoding. Shared structure across individuals already supports cross-subject readout through manifold alignment [9], contrastive embeddings [19], and, on this cohort, atlas-aligned foundation models [21]. Multi-participant pooling can also reduce overfitting: HTNet, for example, pools participant data to avoid fitting a decoder to a small single-subject sample [13], and pooled pretraining across 83 animals followed by fine-tuning on held-out animals reaches state-of-the-art encoding and decoding on a standardized cross-subject cohort [12]. Our question differs in two ways. Primarily, prior work measures accuracy or transfer at a fixed time, while we look at how that accuracy changes over days, to see whether pooled decoders age more slowly. Further, we keep the training-set size fixed with the count-matched control, hence being able to delegate the relevant overfitting not to limited data but to idiosyncratic features that cause drift.

The closest work to ours combines both axes: Meta-AlignNN [10] uses the consistency of neural population dynamics across individuals to build a decoder that remains stable across subjects, time and tasks, and demonstrates it over two years in three monkeys. That work and ours point in the same practical direction, and differ in kind rather than in topic. Meta-AlignNN proposes a method for achieving stability; we measure whether a difference in aging rate exists at all between a decoder trained on the subject and one trained on others, with the training-set size held fixed so that the comparison isolates whose data the decoder saw. A measured effect of this kind is what would justify the design choice such methods already make.

Conservation of cortex-wide dynamics. Cortex-wide activity contains shared, low-dimensional, movement-related components [11, 7, 8], and latent dynamics are preserved across animals performing similar behaviour to the point that a decoder fit to one animal transfers to another [17]. That conservation is the mechanism a pooled decoder would exploit, and it is why pooling is possible at all here. It does not, however, imply that pooling will *resist drift*: a code can be shared across individuals and still move within each of them at the same rate. Whether the shared component is also the more temporally stable one is the empirical question we ask.

3 Data and methods

BraiDyn-BC [6] (DANDI:001425) provides hemodynamically corrected dorsal-cortex $\Delta F/F$ parcellated into 44 Allen Common Coordinate Framework regions for 25 mice, each completing $\sim$15 daily operant sessions over $\sim$3 weeks. For each of the four events we build a 44-dimensional evoked feature representing the mean $\Delta F/F$ over 0–1 s minus the baseline, per parcel, and decode it against matched negatives with ℓ_2-regularized logistic regression, scored by the area under the receiver-operating characteristic curve (AUC).

Sessions are grouped into an early block and a late block. The personal decoder leaves out one early session at a time; training is on the mouse's remaining early sessions and scoring is both on the left-out session and the late block. The pooled decoder trains once on the *other* mice's early blocks and is scored on the same held-out data.

Aging is the accuracy lost as testing moves from early to late. The per-mouse asymmetry is personal aging minus pooled aging, positive when the personal decoder ages more. Significance is a paired sign-flip permutation test across mice, with cluster-bootstrap confidence intervals.

Holding out whole sessions rather than trials is essential, as only the within-subject arm can share a session between training and test. The external validation cohort is Allen Visual Behavior two-photon [1]: containers spanning at least 14 days whose experiments are all in VISp within a single 50 µm depth bin, leaving 23 mice after quality control. Because different animals have no neuron correspondence, the representation that is comparable is a permutation-invariant summary of the population response, giving a 27-dimensional feature regardless of how many cells were segmented.

The decoded contrast is `is_change`, a real image change against a matched catch trial with no change. Appendix C gives both cohorts in full.

4 Results

4.1 A pooled decoder ages more slowly than a personal one

For each mouse we keep the test data fixed (first the mouse's held-out early sessions, then its late block) and change only the source of the training data: either the mouse itself or the other 24 mice. Across all 100 mouse–event pairs the asymmetry is $+0.017$ AUC ($p = 5 \times 10^{-5}$), with the personal decoder aging more in 73% of pairs; the per-event breakdown is in Table 2 of Appendix B, and the two cohorts are summarised together in Table 1.

The effect differs among the four events: it is largest for reward ($+0.029$, $p = 0.0005$) and lick ($+0.020$, $p = 0.002$), marginal for tone ($+0.015$, $p = 0.08$), and absent for lever pull ($+0.003$, $p = 0.37$). While we report this variation, we do not interpret it, as the four events all fall within roughly the same two seconds of a trial. Accordingly, the pooled estimate over all 100 pairs is the claim we defend, and the per-event column should be read as data rather than as four separate findings.

Table 1: The aging asymmetry on the original cohort and on the external validation dataset. Aging is AUC lost from early to late test with training held fixed; asymmetry is personal − pooled aging, positive when the subject's own decoder ages more. Both rows use the same session-held-out protocol and the same count-matched control. p: paired sign-flip permutation. Per-arm aging for BraiDyn-BC is broken out by event in Table 2.

	Personal	Pooled	Asymmetry (95% CI)	p	own ages more
BraiDyn-BC, widefield, 25 mice	–	–	$+0.017$ $[0.010, 0.024]$	5×10^{-5}	73%
Allen VB, two-photon, 23 mice	$+0.024$	$+0.001$	$+0.023$ $[0.010, 0.037]$	0.003	74%

Count-matched control, aging at equal training-event count. BraiDyn-BC: self − one other mouse $= +0.010$; many others age less than one on all four events. Allen VB: self $+0.024$, one $+0.013$, many $+0.003$; self − one $= +0.011$, many − one $= -0.010$.

4.2 Ruling out the alternative explanations

A pooled decoder contains $\sim$24$\times$ more training events than a personal decoder, so its stability could reflect data volume rather than pooling. We test this with a count-matched control (Fig. 1, left). For each held-out early session s of each mouse we set the training-set size to n, the number of events in that mouse's remaining early sessions, and compare three decoders: *self*, trained on those n events; *one*, trained on n events from a single other mouse; and *many*, trained on n events drawn across the other 24 mice. All three are scored on the same held-out session s and the same late block, so they differ only in whose data they were fit on.

For the events where there was an asymmetry, the count-matched ordering reproduces it. The self decoder ages more than a decoder trained on one other mouse by $+0.010$ AUC on average, and by $+0.007$ for lick, $+0.017$ for reward and $+0.006$ for tone individually. As both decoders see the same number of training events, this is not a data-volume artifact.

The diversity ordering is the most robust result in this analysis: *many* ages less than *one* for all four events, including lever pull (-0.026 vs. -0.020). Training on a diverse set of individuals rather than a single other individual therefore provides a smaller, additional benefit that does not depend on which event is decoded.

We next vary the number of training mice, $N \in \{1, 2, 4, 8, 16, 24\}$ (Fig. 1, right). As N increases, pooled aging becomes slightly more favorable for lick (-0.001 to -0.010), reward (-0.004 to -0.010), and tone ($+0.003$ to -0.010); lever pull saturates with a single training mouse and remains near -0.025. The scaling effect is around 0.01 AUC, comparable to the self-versus-one-other gap of Section 4.2. Most of the resistance to drift is already present when training on a single other individual, and additional individuals refine it further. This analysis never lets the decoder see the held-out mouse, so it is unaffected by the protocol issue below.

4.3 Measuring it correctly

Every number reported above holds out whole recording sessions. Repeating the analysis with a trial-level split inflates the personal decoder's within-block accuracy by $+0.013$ (lever pull), $+0.010$ (lick), $+0.009$ (reward) and $+0.000$ AUC (tone), while leaving the pooled decoder untouched. The inflation therefore lands on one side of the difference, and for lever pull this is the entire effect: a significant-looking $+0.014$ ($p = 0.001$) becomes $+0.003$ ($p = 0.37$). Any comparison of a within-subject decoder against a cross-subject one is exposed to this, since only the within-subject arm can share sessions between train and test.

4.4 The effect replicates on an independent dataset

To further test whether the asymmetry is a property of that one cohort, modality and task, we repeat the analysis on the Allen Visual Behavior two-photon dataset [1]: a different laboratory, cellular two-photon rather than mesoscale widefield imaging, a visual change-detection task rather than a cued lever pull, and a 27-dimensional permutation-invariant population summary rather than a 44-parcel atlas feature. The decoded contrast is a stimulus change, so it is much less coupled to reward and licking than anything decoded on BraiDyn-BC.

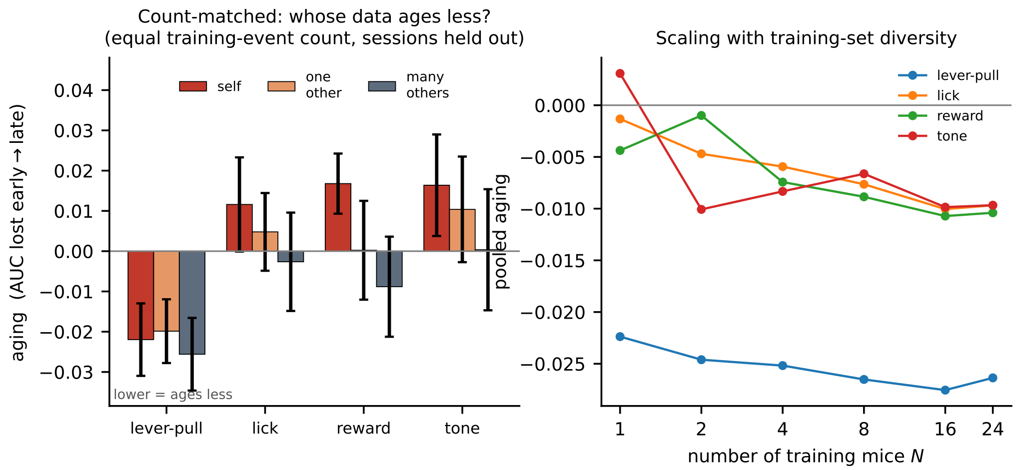

Figure 1: **Left:** count-matched control. With equal numbers of training events, decoder aging (AUC lost from early to late; lower values indicate greater robustness) is compared for the subject's own data (self), one other mouse, and many other mice. **Right:** pooled aging as a function of the number of training mice N. Greater diversity provides a small additional gain, with immediate saturation for lever pull.

The asymmetry reproduces at a comparable magnitude (Table 1). Across 23 mice the personal decoder ages by $+0.024$ AUC and the pooled decoder by $+0.001$, an asymmetry of $+0.023$ $[0.010, 0.037]$, $p = 0.003$, with the mouse's own decoder ageing more in 74% of animals. Early-block accuracy is 0.672 for the personal decoder and 0.638 for the pooled one, so the familiar ordering holds at the start; by the late block the personal decoder leads by only 0.012, about a third of its initial advantage.

The mechanism identified is not identical in both datasets; on BraiDyn-BC the gap narrows mostly because the pooled decoder *gains* over days as the animal becomes more practised (Appendix D), while the personal decoder stays flat. Here the pooled decoder is flat ($+0.001$) and the entire asymmetry comes from the personal decoder losing accuracy. The shared finding within both, hence, is that the personal decoder ages faster.

The count-matched control reproduces as well. At equal training-event count the self decoder ages by $+0.024$, a decoder from one other mouse by $+0.013$ and one drawn across the other mice by $+0.003$. The self-minus-one gap is $+0.011$ $[0.001, 0.022]$, close to the $+0.010$ measured on BraiDyn-BC, though on 23 animals it is marginal by the sign-flip test ($p = 0.054$); self-minus-many is $+0.021$ $[0.010, 0.033]$, $p = 0.002$. The diversity ordering again holds: many other mice age less than one by -0.010 $[-0.016, -0.004]$, $p = 0.003$. The result also does not depend on where the inclusion threshold is drawn, giving $+0.022$ at three or more sessions per mouse (24 mice), $+0.023$ at four (23 mice), $+0.023$ at five (22 mice) and $+0.024$ at six (18 mice).

One aspect of the analysis does not transfer between the two cohorts. Repeating the trial-level split here *lowers* the asymmetry to $+0.014$ ($p = 0.07$) rather than raising it. Separating the two differences on the same trials: trial shuffling still inflates the personal decoder by $+0.015$, as on BraiDyn-BC, but scoring one pooled AUC across sessions instead of averaging per-session values costs -0.030 ($p < 10^{-4}$) and dominates. The corrected estimate is here the *larger* one, so the replication is not an artifact of the split.

5 Discussion

A subject's own decoder overfits idiosyncratic features of its early sessions, some of which later drift. A decoder that has never seen the subject must instead rely on components of the code that are shared across individuals and more stable over time. The count-matched control supports this reading directly: at equal data, other-individual data ages less than the subject's own. Population structure is therefore not only a resource for cross-subject transfer but also for temporal robustness, and the early-week accuracy cost of cross-subject decoding is recovered over time as the pooled decoder remains stable. Unlike within-subject recalibration, this requires no newly labelled data from the individual at test time – which is the practical point, since obtaining that data is the expensive part of deploying a decoder over months.

The effect is small in absolute terms, and we think it is worth being explicit about why we believe it anyway. It rests on 25 animals and 357 recording sessions, which is a large cohort for a longitudinal cortex-wide study, and it is supported by five checks that are not variations on one analysis. A two-block contrast and a day-by-day regression over all fifteen sessions are different estimators and select the same events. The count-matched control fixes the training-set size, so the result cannot be a data-volume artifact. Replacing the linear readout with a 1-D CNN or a GRU leaves it in place, so it is not an artifact of a weak decoder. And holding out whole sessions rather than trials removes an inflation large enough to eliminate one of the four events entirely – the effect reported here is what remains after that correction, not before it. And it reproduced in a second cohort recorded in another laboratory with a different instrument and a different task. Taken together, each of these checks could independently have removed the result, and none of them did.

Code and data availability. Both cohorts are public (DANDI:001425; Allen Brain Observatory). The analysis code and streaming feature pipeline are released.

Limitations. The clearest limitation is what the four events are. They are not four independent behaviors but four thresholded channels of one cued lever-pull trial, overlapping to the point where 100% of reward windows also contain licking, and differing tenfold in event count. We therefore cannot always say which behavior the effect belongs to, nor can we exclude the idea that the per-event ranking is a tracker of how much data each decoder had to overfit. A design with temporally separated, count-matched events would settle both, and is the experiment we would run next.

We measure at the mesoscale, using parcel-averaged activity, and while the two-photon replication does reach single neurons it does so only at one depth in one visual area.

The analysis covers a two-week interval in one species, and nothing here establishes how the asymmetry behaves over the months-to-years horizon on which a deployed decoder has to survive.

Finally, the trial-level inflation of Section 4.3 is worth stating as a caution rather than as a limitation of this work, because it is a limitation we do not have. It is large enough to manufacture an effect that is not there – on lever pull it manufactured the entire effect – and every number we report is measured after holding out whole sessions, not before. We raise it because any comparison of this shape is exposed to it, and the correction is the one in Appendix A.

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A Best-practice checklist

B Per-event results

C Data and methods in full

C.1 Datasets and preprocessing

BraiDyn-BC [6] (DANDI:001425, CC-BY) provides, for each mouse and session, hemodynamically corrected dorsal-cortex $\Delta F/F$ traces parcellated into 44 Allen Common Coordinate Framework regions at $\sim$30 Hz, together with time-aligned behavioral channels. The dataset includes 25 mice, each of which completed $\sim$15 daily operant sessions over $\sim$3 weeks. We stream the 44-region traces and behavioral channels without downloading the raw movies.

The two cohorts used in this paper were built for different purposes, and the ways they differ are what make the comparison informative rather than redundant. BraiDyn-BC is densely longitudinal, with roughly fifteen sessions per mouse over three weeks, and it supplies a shared coordinate system:

Best practice for longitudinal within- versus cross-subject comparisons

1. **Hold out whole recording sessions, not trials.** Only the within-subject arm can share a session between train and test, so a trial-level split inflates that arm alone. Here it inflated within-block accuracy by $+0.013$ (lever pull), $+0.010$ (lick), $+0.009$ (reward) and $+0.000$ AUC (tone), and on lever pull it accounted for the entire apparent effect: $+0.014$ ($p = 0.001$) became $+0.003$ ($p = 0.37$).
2. **Count-match the training sets before comparing arms.** A pooled decoder contains $\sim 24\times$ more training events than a personal one. Fix the training-set size to n and fit *self*, *one* and *many* on the same n, scored on the same held-out session.
3. **Never score an arm on the block it was fit on.** In the count-matched control the natural implementation scores the self decoder on its own training block – not a cross-validated quantity at all, and enough to reverse the ordering the control exists to establish.
4. **Report each arm's aging separately, not only the difference.** The asymmetry is a difference of two quantities that can move for different reasons: on BraiDyn-BC it narrows because the pooled decoder gains, on Allen because the personal decoder loses.
5. **Average per-session scores rather than pooling one score across sessions.** On the Allen cohort, scoring a single pooled AUC across sessions instead of averaging per-session values cost -0.030 ($p < 10^{-4}$) and dominated the leakage term it was confounded with.

Table 2: Personal vs. pooled aging across days (AUC lost from early to late test, holding training fixed), per event, $n = 25$ mice, with whole early sessions held out. Asymmetry = personal − pooled aging; a positive value means the mouse's own decoder ages more. p: paired sign-flip permutation. The rightmost column gives the same asymmetry under a trial-level split, the protocol this analysis replaces (Section 4.3).

Event	Personal	Pooled	Asymmetry (95% CI)	p	own ages more	trial-split
Lever-pull	-0.022	-0.025	$+0.003$ $[-0.004, 0.011]$	0.37	64%	$+0.014$
Lick	$+0.012$	-0.009	$+0.020$ $[0.008, 0.032]$	0.0023	80%	$+0.028$
Reward	$+0.017$	-0.012	$+0.029$ $[0.014, 0.046]$	0.0005	80%	$+0.032$
Tone	$+0.016$	$+0.001$	$+0.015$ $[0.001, 0.032]$	0.077	68%	$+0.022$
Pooled			$\mathbf{+0.017}$ $[0.010, 0.024]$	$\mathbf{5 \times 10^{-5}}$	**73%**	$+0.024$

the 44 Allen parcels mean that region k in one mouse is the same region in every other, so a pooled decoder can be fit directly in a common feature space. The Allen Visual Behavior cohort has neither property. Its longitudinal subset is sparser, and because it is cellular two-photon imaging there is no correspondence between animals at all – one mouse's cell 17 is unrelated to another's – so a common feature space has to be constructed rather than assumed. The recording modality, the task, and the decoded contrast all differ as well. A result that survives both is therefore not a property of one atlas, one instrument, or one behavior.

C.2 Features and decoders

We detect the onset of four events (lever pull, lick, reward, and tone) as rising edges, with a 1 s refractory period. For each onset we construct a 44-dimensional evoked feature by subtracting the mean pre-onset baseline (-0.5–0 s) from the mean post-onset $\Delta F/F$ response (0–1 s) in each parcel. Negative samples are drawn from matched-count windows more than 2 s from any onset *of the same event*; they are not screened against the other three events. Every classification is a binary decode of one event against these matched negatives, scored by area under the ROC curve (AUC). The four decoders are therefore four independent binary detectors, not a four-way classifier: nothing in this paper asks a model to distinguish one event from another.

The primary decoder is a standardized ℓ_2-regularized logistic regression on the 44-dimensional evoked feature. This is a linear readout of the parcel-space code: it recovers the component of behavior that is linearly present in the population average, and it is the natural comparison to prior cross-subject decoders, which are overwhelmingly linear [9, 13]. Every condition below uses this same readout, so the only variable is which data the decoder was trained on.[1]

C.3 Temporal blocks, aging, and the session-held-out protocol

We group each mouse's sessions into an early block (days 1–5) and a late block (days 11–15) and report AUC under four conditions. The personal decoder is estimated by leaving out one whole early *session* at a time: for each early session s we train on the mouse's remaining early sessions and score the resulting decoder both on s (within-mouse same-block, WS) and on the late block (within-mouse across-block, WX), then average over s. Because WS and WX come from the same fitted decoders, they differ only in which data they are tested on. The pooled decoder is trained once on the other mice's early blocks and scored on the same held-out sessions s (LS) and on the late block (LX).

Holding out sessions rather than trials is essential and not a detail. Trials within one recording share illumination and hemodynamic state, slow arousal drift, and temporal autocorrelation, so a trial-level split lets the personal decoder train and test inside the same session. The pooled decoder is trained on other animals entirely and cannot benefit this way, so the inflation is one-sided and lands directly on the quantity of interest; Section 4.3 reports what it costs per event.

Aging is the accuracy lost when testing moves from the held-out mouse's early data to its late block while training stays fixed: personal aging is $\mathrm{WS} - \mathrm{WX}$ and pooled aging is $\mathrm{LS} - \mathrm{LX}$. The per-mouse aging asymmetry is $(\mathrm{WS} - \mathrm{WX}) - (\mathrm{LS} - \mathrm{LX})$; a positive value means the personal decoder ages more. We assess significance with a paired sign-flip permutation test across mice (20,000 permutations) and compute 95% confidence intervals with a cluster bootstrap over mice.

C.4 The external validation cohort

Section 4.4 repeats the analysis on the Allen Visual Behavior two-photon dataset [1]. We select containers spanning at least 14 days whose experiments are all in VISp within a single 50 µm depth bin, which is the cross-subject correspondence available here; the largest such band, 150–199 µm, holds 25 mice and 184 experiments. We require at least 15 segmented cells and 40 usable trials per experiment, leaving 166 experiments and 23 mice with at least four sessions. Sessions are streamed from S3 rather than downloaded.

As different animals have no neuron correspondence and the longitudinal subset is single-plane, averaging over the field of view would give a one-dimensional feature. The representation that is comparable across animals is therefore a permutation-invariant summary of the population response: per neuron we take the mean $\Delta F/F$ over 0–1 s minus the mean over -0.5–0 s, then describe the distribution of those evoked responses by 25 quantiles plus its mean and standard deviation, giving a fixed 27-dimensional feature regardless of how many cells were segmented. The decoded contrast is `is_change`, a real image change against a matched catch trial with no change, which is a stimulus contrast rather than a reward- or lick-coupled one. Each mouse's sessions are split by date into early and late halves, and the personal decoder, the pooled decoder, aging, the asymmetry and the count-matched control are all defined exactly as above.

D Day-by-day decay regression

The block split compares two windows, but the effect should also appear as a continuous trend across all fifteen days. For each session day we compute the accuracy of the personal decoder and the pooled decoder and regress each on day; on early days the personal decoder is trained by leave-one-session-out, so it is never tested on a session it trained on. The personal decoder starts ahead, leading by about 0.035 AUC on day 1 as expected for a decoder trained on the same animal. That lead then erodes: the gap narrows by 0.0014 AUC per day ($p = 5 \times 10^{-5}$; narrowing in 65% of pairs), so by day 15 roughly 60% of the early-week advantage is gone.

[1] The result is not a property of the linear readout. Repeating the analysis with a 1-D convolutional network over time and a GRU over the frame sequence (full evoked window, -0.5 to $+1.0$ s, 45×44), under the same session-held-out protocol, the asymmetry survives for both events that carry it: lick gives $+0.020$, $+0.012$ and $+0.010$ and reward $+0.029$, $+0.015$ and $+0.012$ under the linear, CNN and GRU readouts, all significant. Under a trial-level split the nonlinear estimates appear *larger* instead, which is the inflation of Section 4.3: a higher-capacity model exploits it more effectively than a linear one.

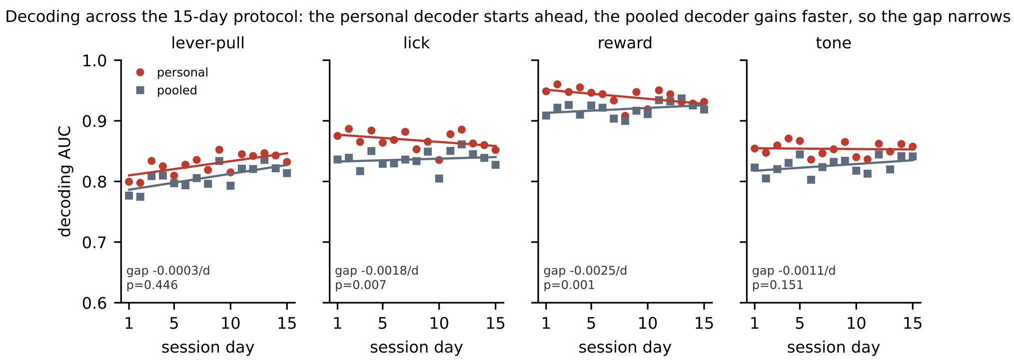

Figure 2: Decoding accuracy across the 15-day protocol for the personal decoder (own early data) and the pooled decoder (other mice's early data), with per-day points and linear fits. The personal decoder starts ahead, and the pooled decoder gains more over days, so the gap between them narrows. The narrowing is significant for lick and reward, the two events where the personal decoder also loses accuracy over days, and is not significant for lever pull or tone. The per-event gap slope and its p (paired sign-flip over mice) are shown.

Averaged over the four events, the pooled decoder gains accuracy over days ($+0.0015$ AUC/day) as the sessions become easier – the mouse is more practiced – while the personal decoder stays flat ($+0.0000$ AUC/day). The personal decoder, fit to idiosyncrasies, is unable to ride that wave, and the lift it fails to take is what opens the gap. Per event, the personal decoder does lose accuracy for lick (-0.0010 AUC/day) and reward (-0.0016 AUC/day), and these are the two events where the narrowing is individually significant ($p = 0.007$ and $p = 0.001$; Fig. 2).